State diagram exercise:

Try to create one that recognizes <u>comments</u> in C or C++ source code.

Careful of things like this:

```
string s = "/* I'm not a comment. */";
```

---

New topic: pointers + dynamic memory management.

<u>Pointers</u>

Q: what are they?

A: variables that store a <u>memory address</u>

Q: Why?

A: You'll see soon...

Q: How to declare?

A: `int* p;` // p can hold the

Aside: int* p ≡ int *p
≡ int*p

// address of an integer.

Q: How to use?? Need to get an address...

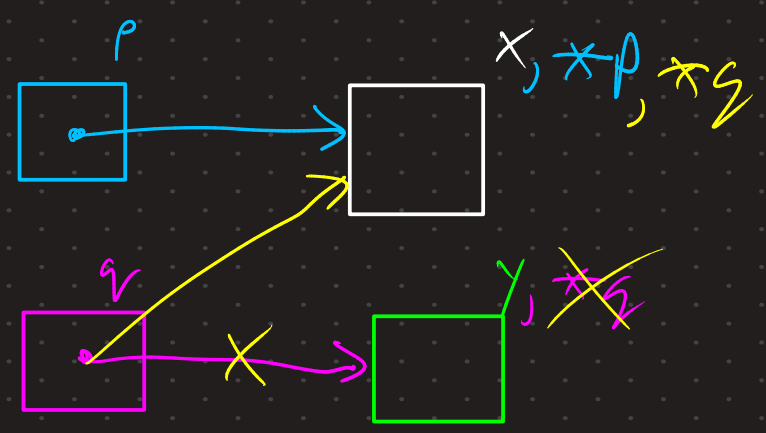
A: Say we had int x;.
then p = &x; will work.

"address of" operator.

There's more: how to change memory contents through a pointer?

*p = 100;

"dereference" operator

In pictures:

```
int x;
int *p = &x;
int y;
int *q = &y;
q = p;
```